

IF



What does it do?

It checks whether a condition is true or not and on the basis of that returns a value

Syntax

=IF(Logical_Test, [value_if_true], [value_if_false])

Logical_Test

This will be the condition which you want to check, so greater than, less than or equal to signs can be used for numbers. When you want to evaluate a condition based on **text value**, then use double quotes (") with equal to signs (Row 27 for example)

Value_if_True

If the logical test holds, then excel will return this value

Value_if_False

If the logical test does not hold, then excel will return this value

Example

Using IF formula to identify if a student has passed an exam or not

Student ID	Marks	Pass or Fail	Formula
S001	90	Pass	=IF(B22>=40,"Pass","Fail")
S002	70	Pass	=IF(B23>=40,"Pass","Fail")
S003	20	Fail	=IF(B24>=40,"Pass","Fail")
S004	41	Pass	=IF(B25>=40,"Pass","Fail")

Note

If instead of marks, we had Grades A, B, C i.e. **text values** then the above formulas would have changed by IF(B21="A","Pass","Fail")

Exercise

Write a formula using IF, Salary for an AVP will be \$2,000 and for an analyst will be \$1,000

Emp ID	Designation	Salary
E001	AVP	
E002	Analyst	
E003	Analyst	
E004	Analyst	

AND



What does it do?

Checks if all arguments are true and returns TRUE value else returns FALSE value. You need to have minimum 2 conditions to check, AND function is used with IF condition a lot

Syntax

=AND(Logical 1, Logical 2, Logical 3, and so on...)

Logical 1

First condition which needs to be checked

Logical 2

Second condition which needs to be checked

Example 1

Testing condition 1 and condition 2 to check the numbers are greater than 0 using AND

Condition 1	Condition 2	Result	Formula
1	2	TRUE	=AND(A19>0,B19>0)
1	-2	FALSE	=AND(A20>0,B20>0)

Example 2

Use IF with AND function to check if student has passed or not. If the student has failed in 1 of the 2 subjects, then the student will be considered as Failed

Student ID	Subject 1	Subject 2	Overall Result	Formula
S001	PASS	PASS	PASS	=IF(AND(B26="PASS",C26="PASS"),"PASS","FAIL")
S002	PASS	FAIL	FAIL	=IF(AND(B27="PASS",C27="PASS"),"PASS","FAIL")
S003	FAIL	PASS	FAIL	=IF(AND(B28="PASS",C28="PASS"),"PASS","FAIL")
S004	PASS	PASS	PASS	=IF(AND(B29="PASS",C29="PASS"),"PASS","FAIL")

Exercise

Write a formula using AND with IF to find if an employee should get a "GOOD" increment or not. Sales made by the employee should be more than 50 and he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)	Formula
E001	AVP	51	YES	YES	=IF(AND(C36>50,D36="YES"),"YES","NO")
E002	Analyst	100	NO	NO	=IF(AND(C37>50,D37="YES"),"YES","NO")
E003	Analyst	49	YES	NO	=IF(AND(C38>50,D38="YES"),"YES","NO")
E004	Analyst	75	YES	YES	=IF(AND(C39>50,D39="YES"),"YES","NO")

Nested IF's

What does it do?

Nested IF conditions help in checking multiple conditions together by using IF conditions within IF condition.

Syntax

`=IF(Logical_Test, [value_if_true], IF(Logical_Test, [value_if_true], [value_if_false]))`

Logical_Test

This will be the condition which you want to check, so greater than, less than or equal to signs can be used for numbers. When you want to evaluate a condition based on **text value**, then use double quotes (") with equal to signs

Value_if_True

If the logical test holds, then excel will return this value

Value_if_False

If the logical test does not hold, then excel will return this value

Nested IF

From Excel 2007 onwards, we can use 64 IF conditions in one formula whereas Excel 2003 allows for only 7 IF conditions

Example

Use Nested IF to put the grade a student gets in an exam in Table B. Score Ranges for different grades are given in Table A.

TABLE A		TABLE B			
Score	Grade	Student ID	Marks	Grade	Formula
>70	A	S001	90	A	=IF(E26<41,"C",IF(E26<71,"B","A"))
41-70	B	S002	70	B	=IF(E27<41,"C",IF(E27<71,"B","A"))
<40	C	S003	20	C	=IF(E28<41,"C",IF(E28<71,"B","A"))
		S004	41	B	=IF(E29<41,"C",IF(E29<71,"B","A"))

Exercise

Write a formula using Nested IF conditions to check what level increment an employee should get and then convert the amount into dollars using Format

TABLE A		TABLE B				Formula
Performance	Increment	Emp ID	Designation	Performance	Result	
A	\$1,000	E001	AVP	A	\$1,000	=IF(F36="A",1000,IF(F36="B",500,250))
B	\$500	E002	Analyst	A	\$1,000	=IF(F37="A",1000,IF(F37="B",500,250))
C	\$250	E003	Analyst	B	\$500	=IF(F38="A",1000,IF(F38="B",500,250))
		E004	Analyst	C	\$250	=IF(F39="A",1000,IF(F39="B",500,250))

OR



What does it do?

Checks if any of the conditions are true or not. It returns TRUE value even if one of the conditions holds else returns FALSE value. Basically, for a FALSE to appear, no conditions should hold. You need to have minimum 2 conditions to check, AND function is used with IF condition a lot

Syntax

`=OR(Logical 1, Logical 2, Logical 3, and so on...)`

Logical 1

First condition which needs to be checked

Logical 2

Second condition which needs to be checked

Example 1

Testing condition 1 and condition 2 to check the numbers are greater than 0 using OR

Condition 1	Condition 2	Result	Formula
1	2	TRUE	=OR(A20>0,B20>0)
1	-2	TRUE	=OR(A21>0,B21>0)
-1	-2	FALSE	=OR(A22>0,B22>0)

Example 2

Use IF with OR function to check if student has passed or not. If the student has failed in 1 of the 2 subjects, then the student will be considered as Failed

Student ID	Subject 1	Subject 2	Overall Result	Formula
S001	PASS	PASS	PASS	=IF(OR(B28="FAIL",C28="FAIL"),"FAIL","PASS")
S002	PASS	FAIL	FAIL	=IF(OR(B29="FAIL",C29="FAIL"),"FAIL","PASS")
S003	FAIL	PASS	FAIL	=IF(OR(B30="FAIL",C30="FAIL"),"FAIL","PASS")
S004	PASS	PASS	PASS	=IF(OR(B31="FAIL",C31="FAIL"),"FAIL","PASS")

Exercise

Write a formula using OR with IF to find if an employee should get a "GOOD" increment or not. Either Sales made by the employee should be more than 50 or he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)	Formula
E001	AVP	51	YES	YES	=IF(OR(C38>50,D38="YES"),"YES","NO")
E002	Analyst	49	NO	NO	=IF(OR(C39>50,D39="YES"),"YES","NO")
E003	Analyst	49	YES	YES	=IF(OR(C40>50,D40="YES"),"YES","NO")
E004	Analyst	75	YES	YES	=IF(OR(C41>50,D41="YES"),"YES","NO")